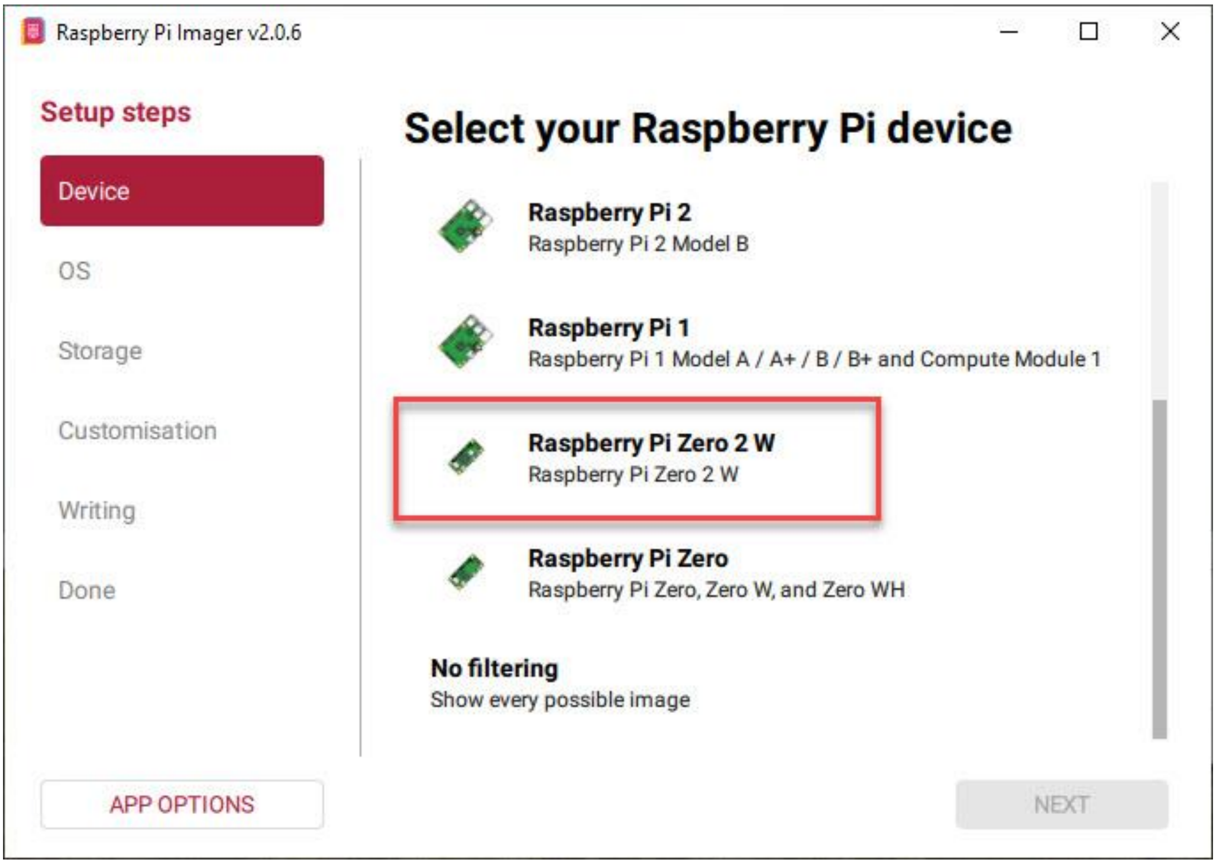
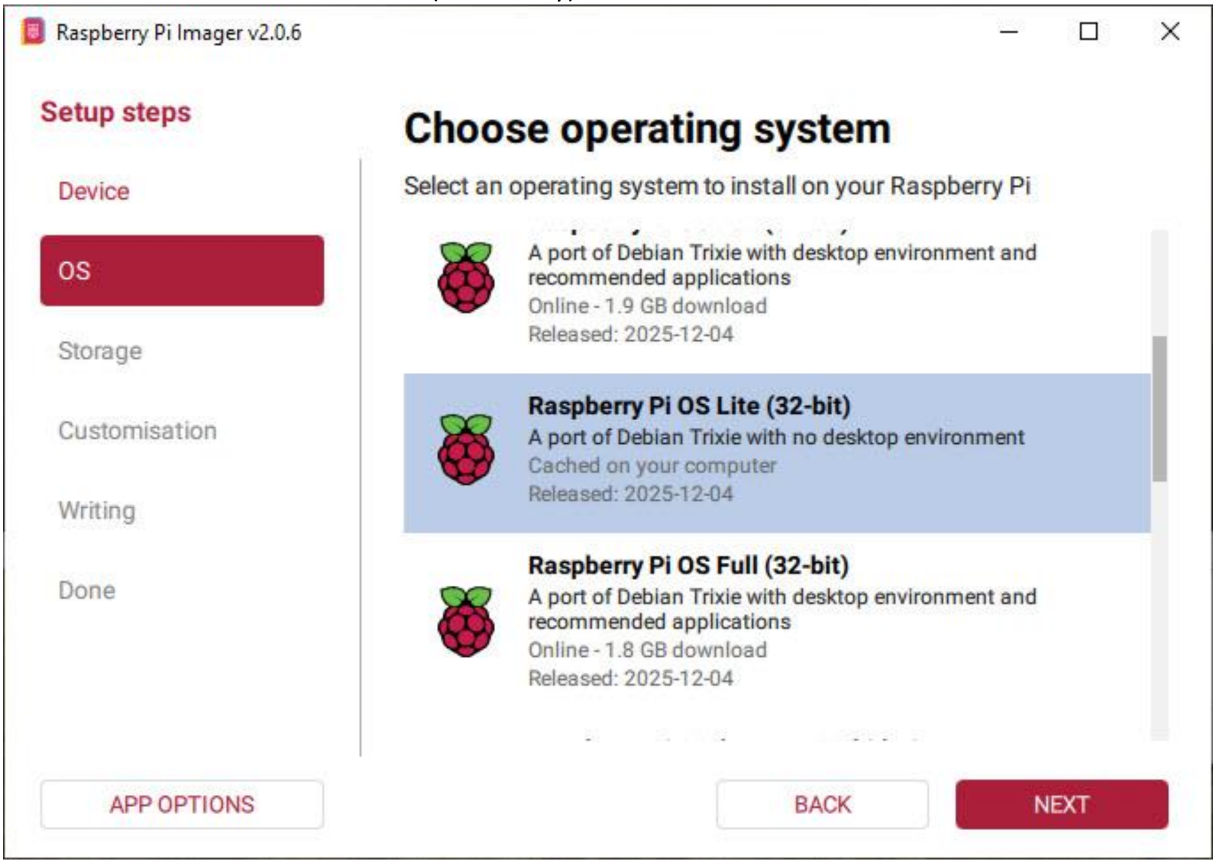
Pi-Ager install on Raspberry Pi 3/Pi 4/Pi 5/Pi zero (2)w under Pi OS 12 Lite Bookworm/Trixie

- For all Pi:
  Download and install Raspberry Pi Imager v2.0.6 or later from https://www.raspberrypi.com/software/.

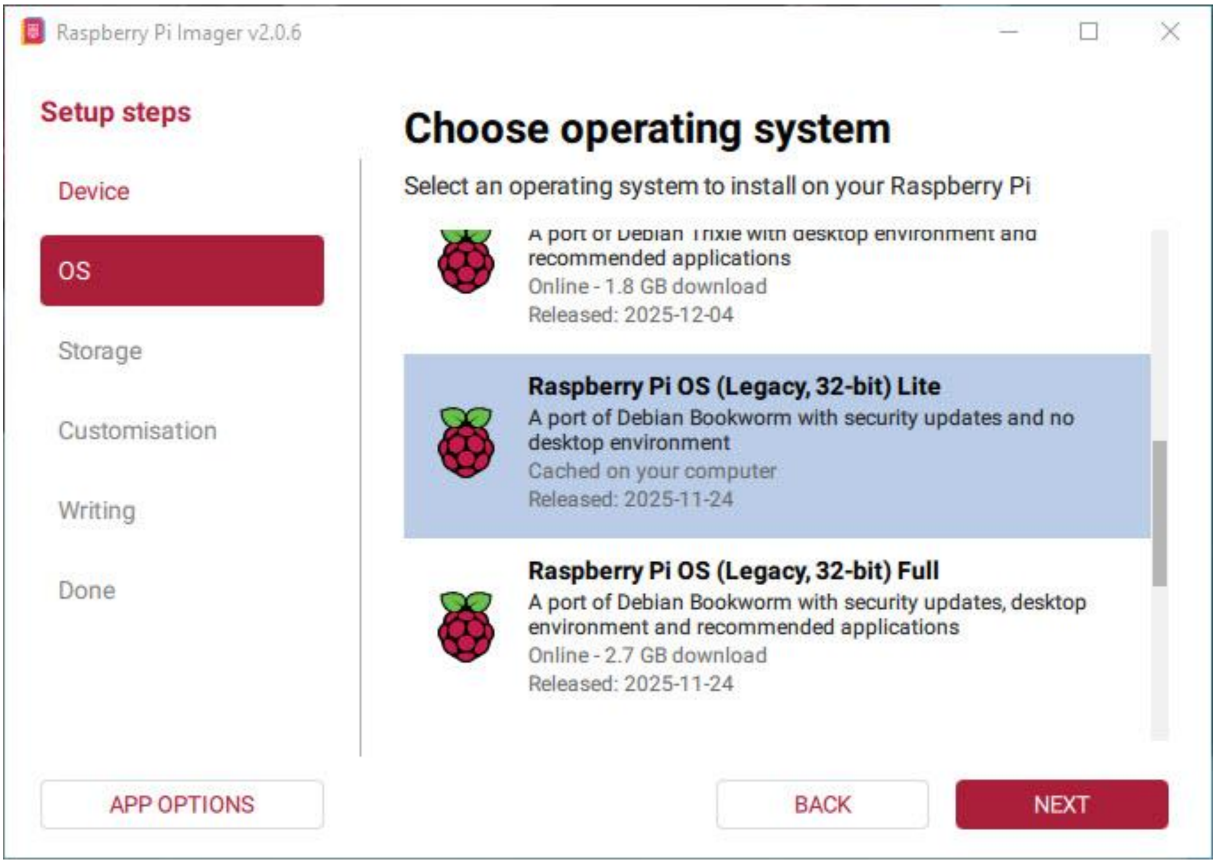
  Start Raspberry Pi Imager, then continue with the follow steps:



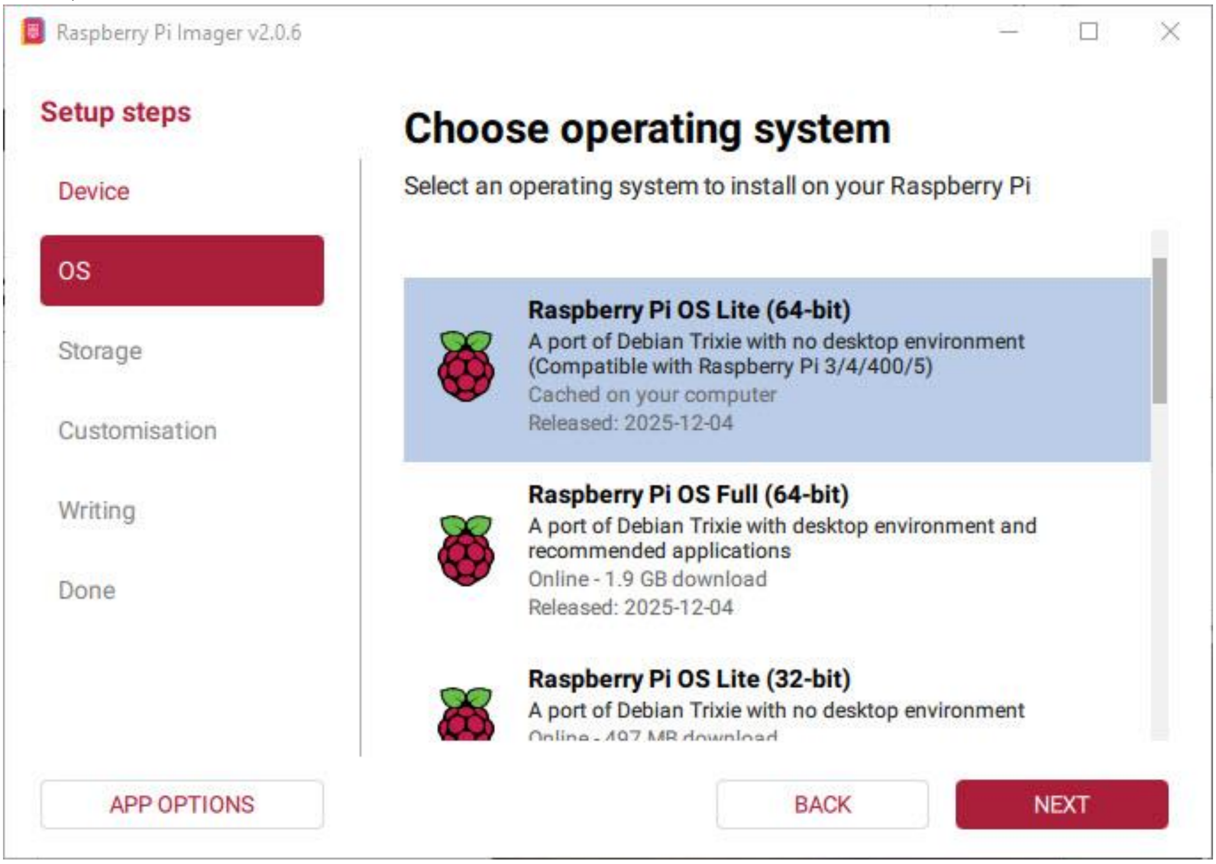
Select Pi-OS 32 Bit Trixie for Pi Zero (32-bit only)

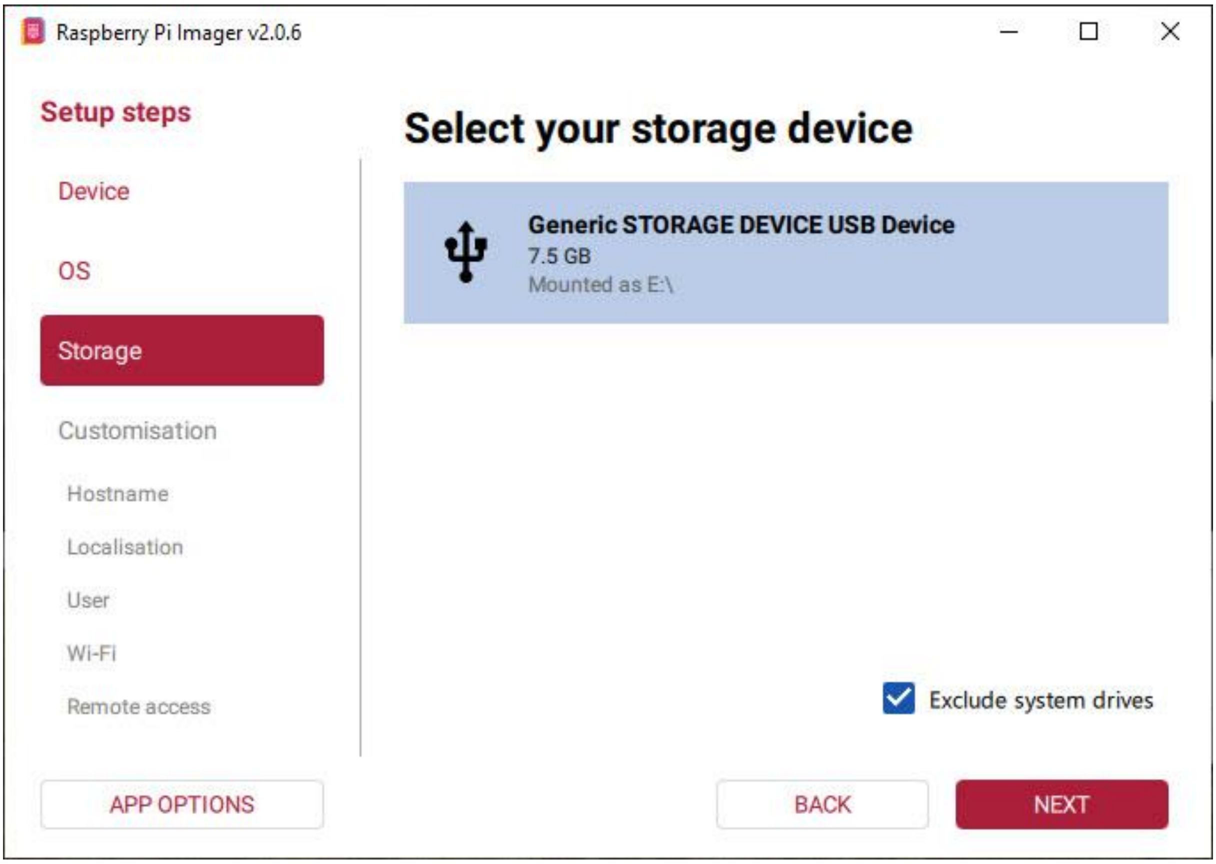


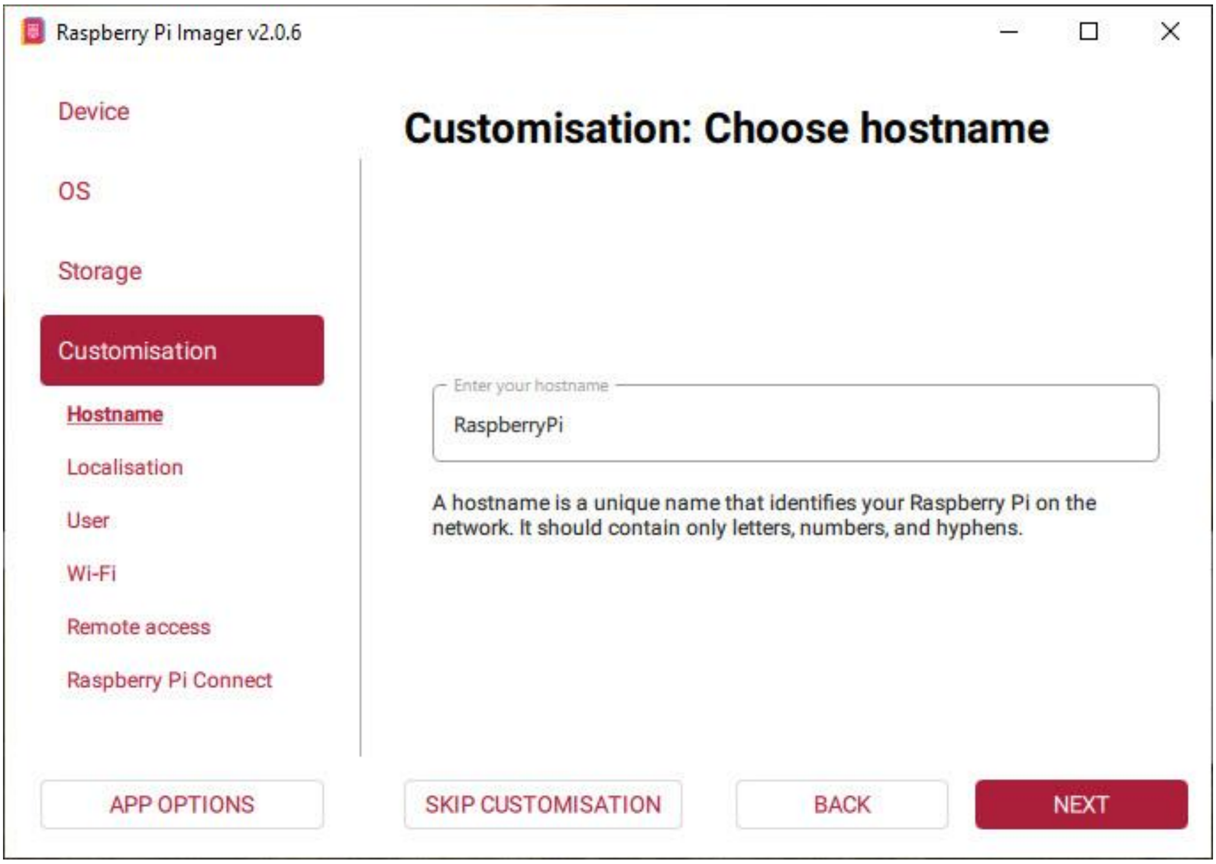
Or select Pi-OS 32 Bit Bookworm:

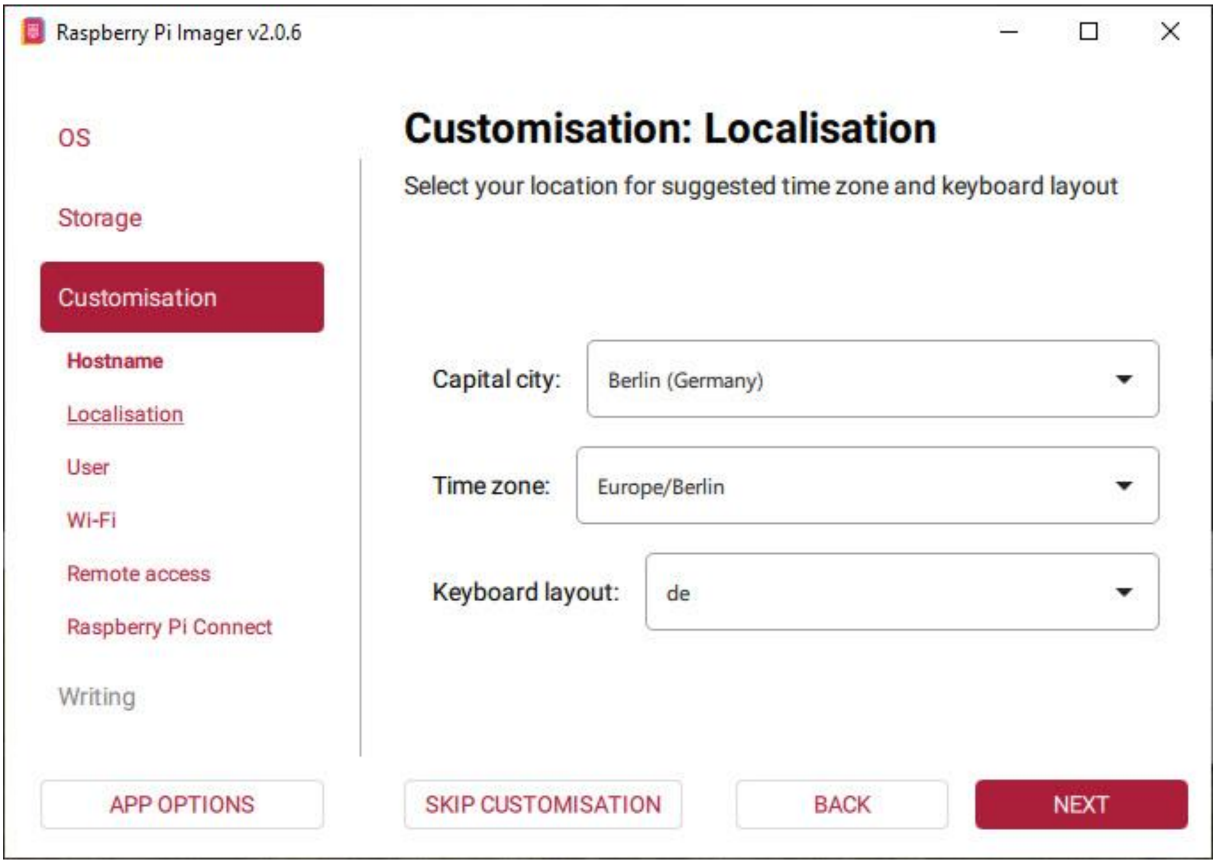


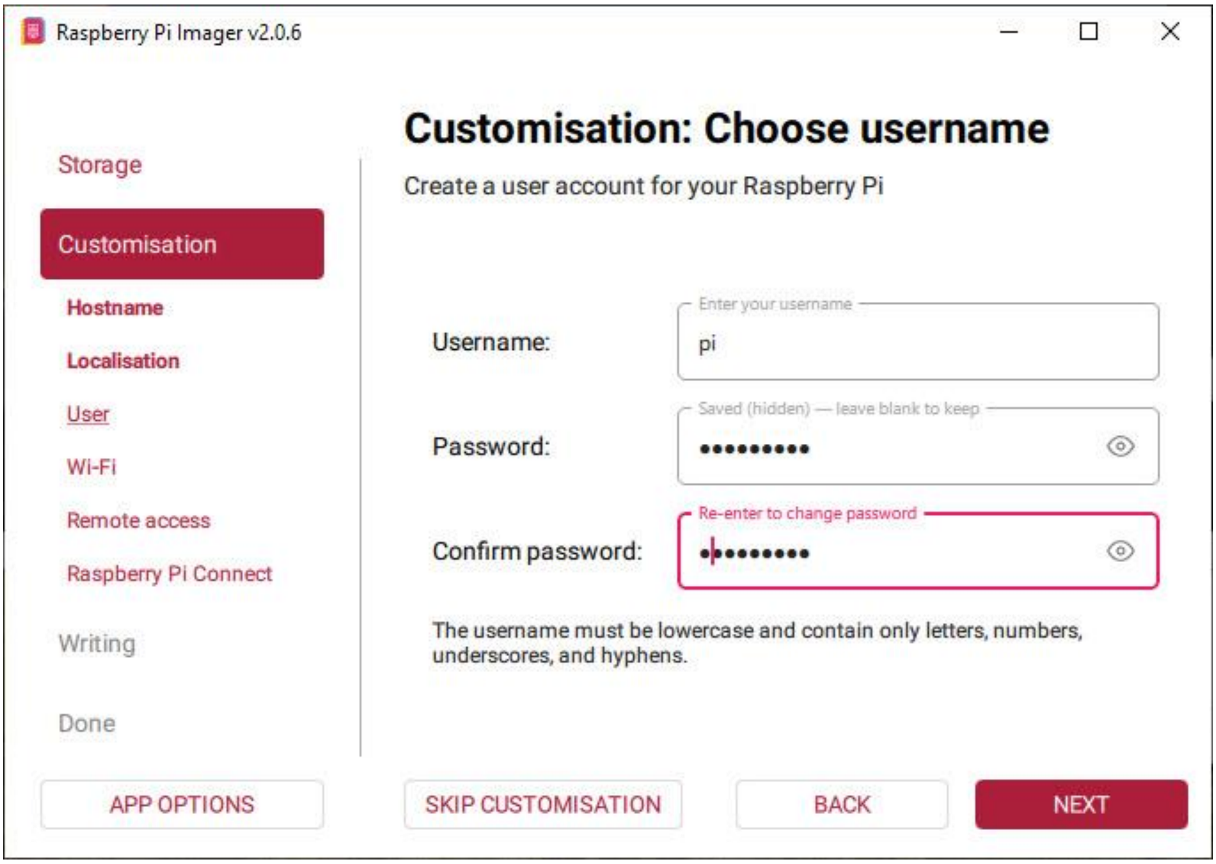
Or if you want to install pi-ager on Raspberry Pi-5 that needs a 64 Bit PI-OS, select Trixie.
Pi-4B, Pi-zero 2W also can use 64-bit Trixie.

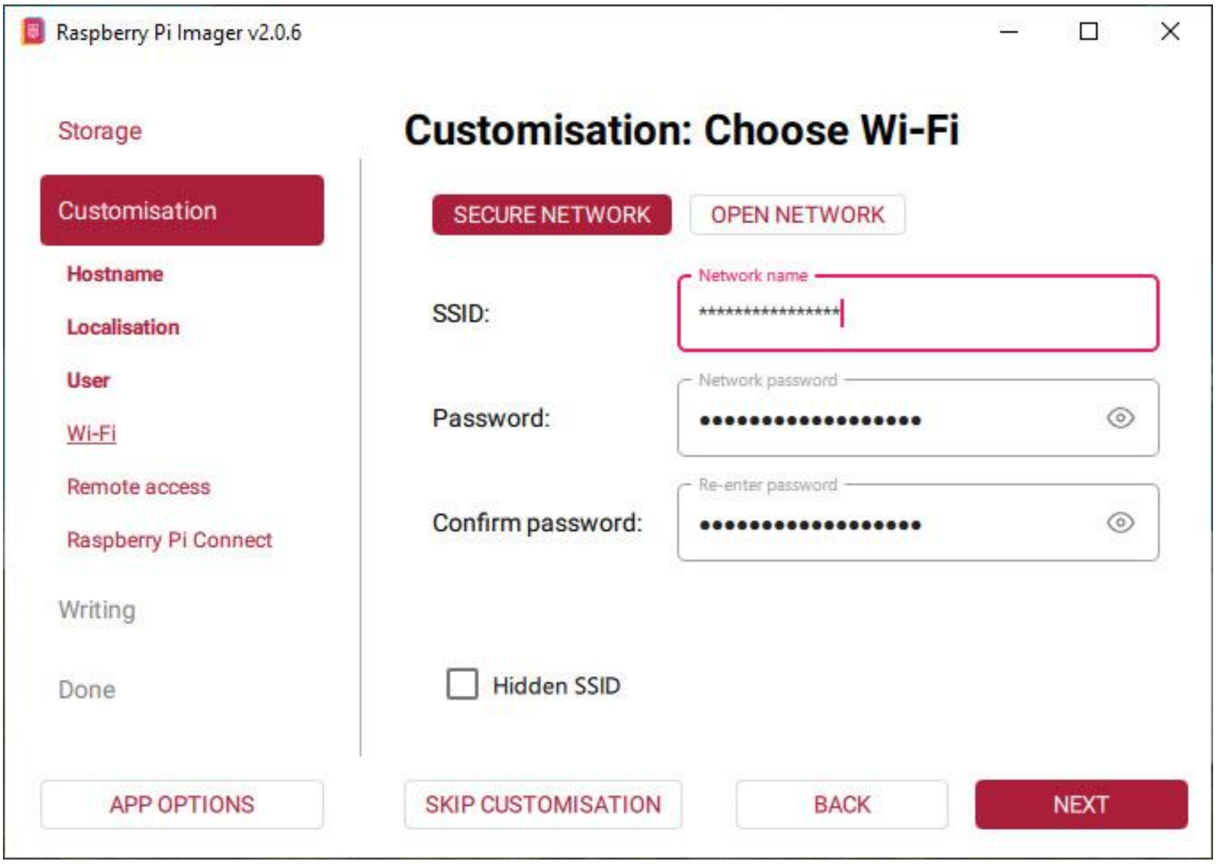


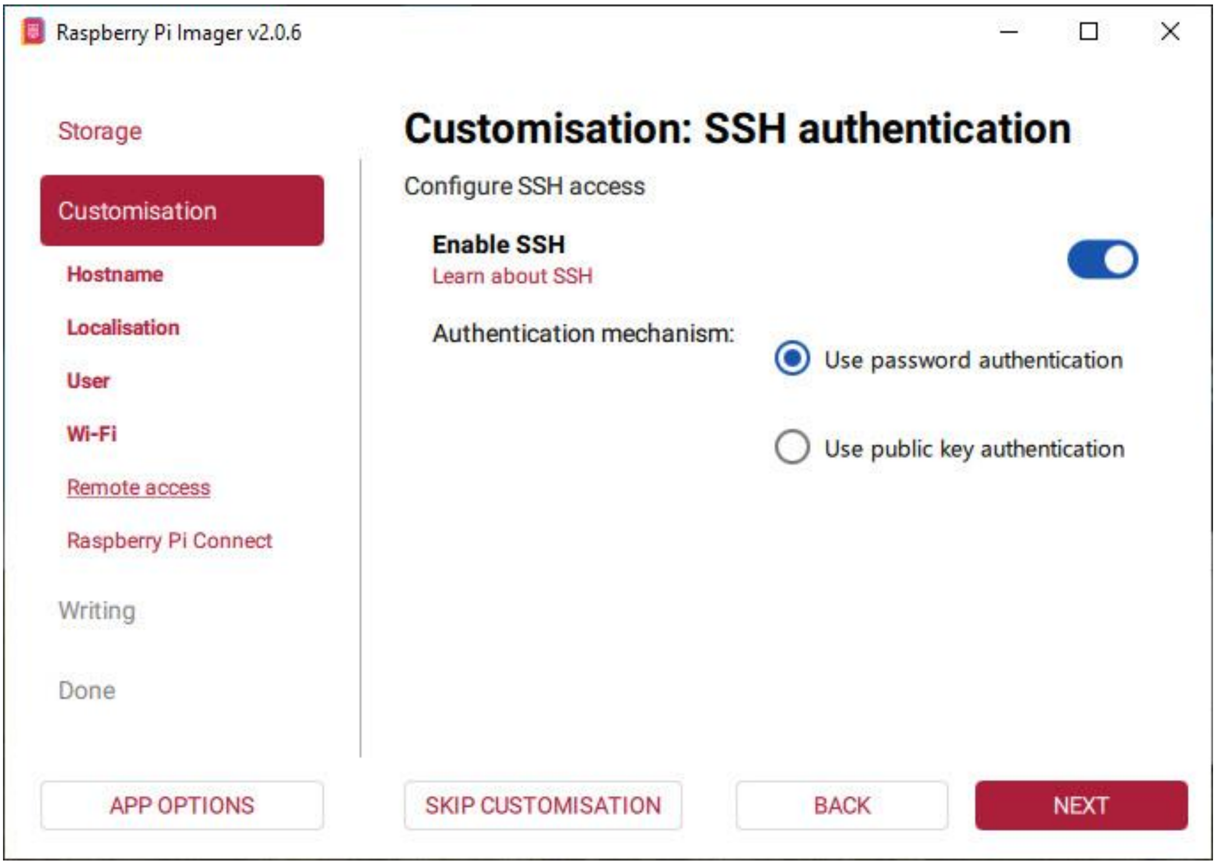


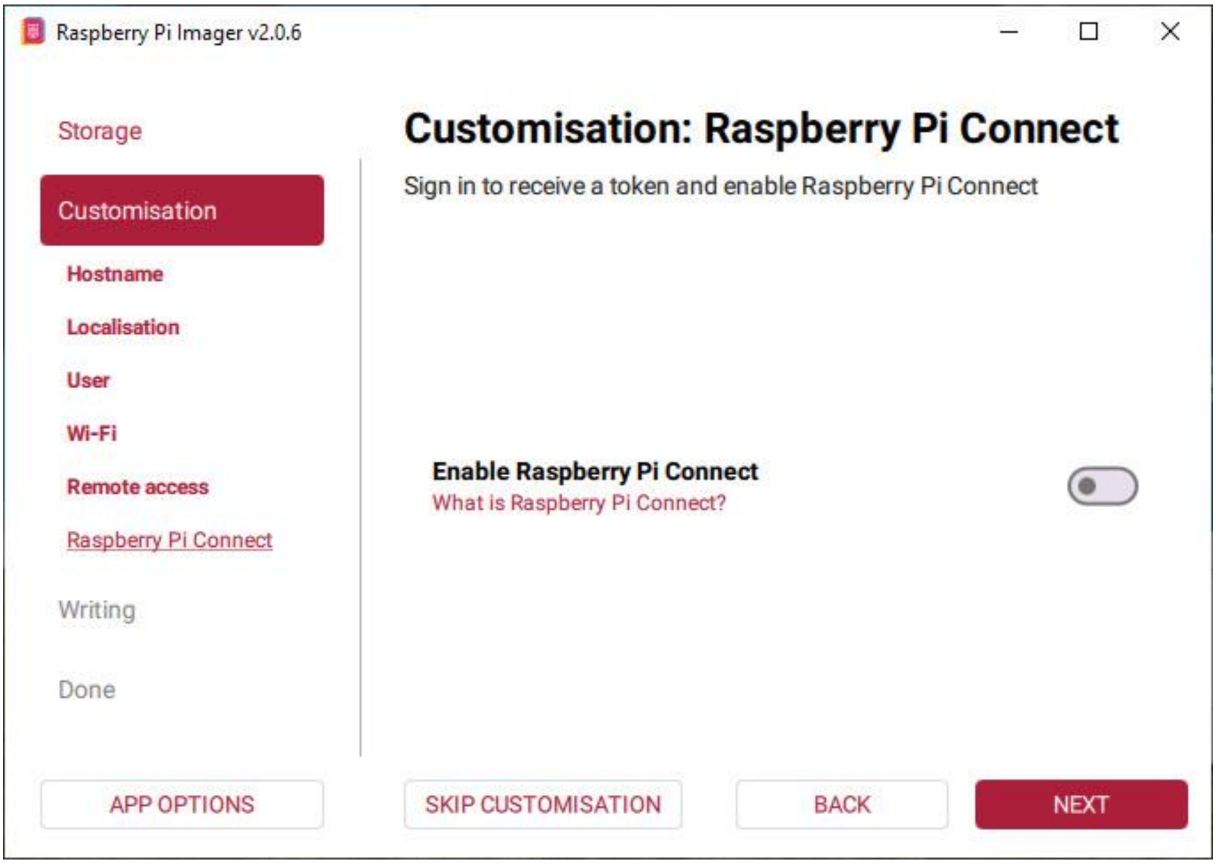

Raspberry Pi Imager v2.0.6
Device
OS
Storage
Customisation
Hostname
Localisation
User
Wi-Fi
Remote access
Raspberry Pi Connect
Customisation: Choose hostname
Enter your hostname
RaspberryPi
A hostname is a unique name that identifies your Raspberry Pi on the network. It should contain only letters, numbers, and hyphens.
APP OPTIONS
SKIP CUSTOMISATION
BACK
NEXT

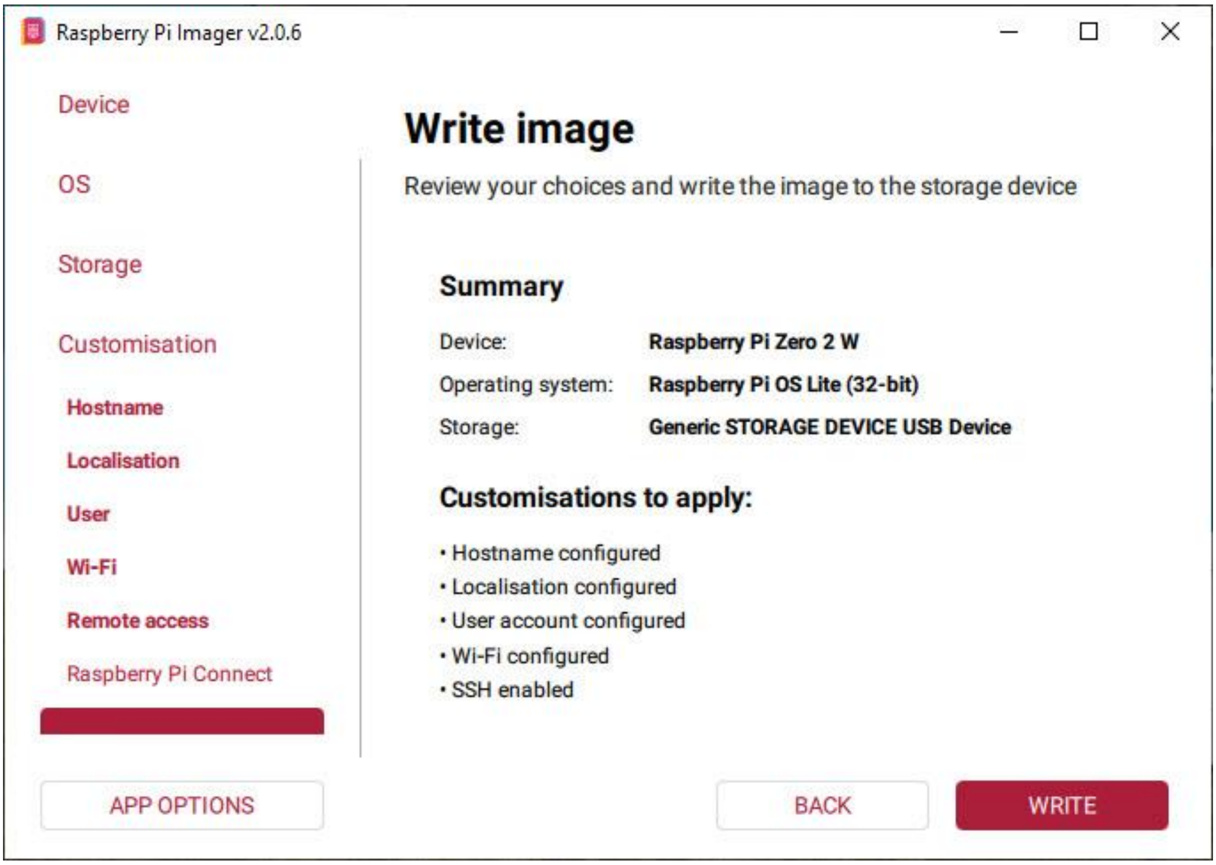

Raspberry Pi Imager v2.0.6
OS
Storage
Customisation
Hostname
Localisation
User
Wi-Fi
Remote access
Raspberry Pi Connect
Writing
Customisation: Localisation
Select your location for suggested time zone and keyboard layout
Capital city:
Berlin (Germany)
Time zone:
Europe/Berlin
Keyboard layout:
de
APP OPTIONS
SKIP CUSTOMISATION
BACK
NEXT

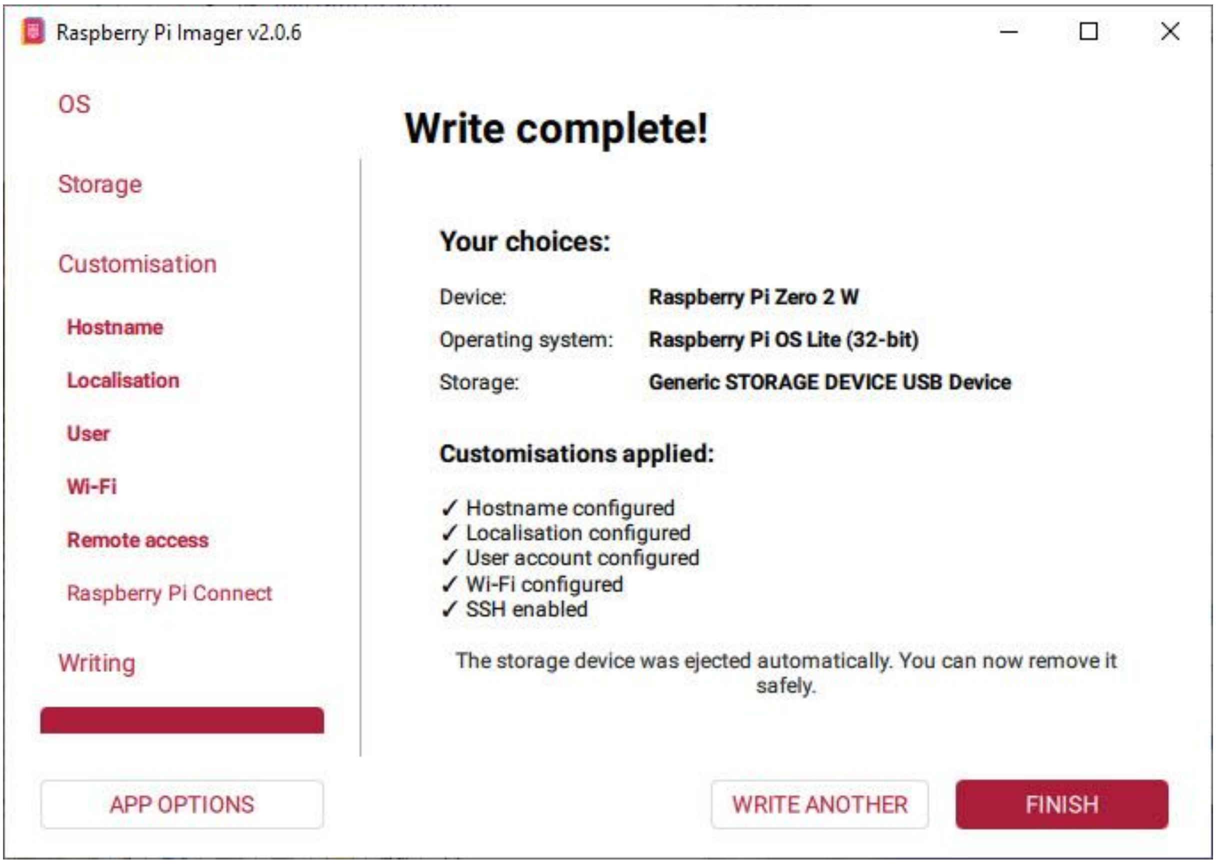

Raspberry Pi Imager v2.0.6
Customisation: Choose username
Create a user account for your Raspberry Pi
Storage
Customisation
Hostname
Localisation
User
Wi-Fi
Remote access
Raspberry Pi Connect
Writing
Done
Username:
Enter your username
pi
Password:
Saved (hidden) — leave blank to keep
Confirm password:
Re-enter to change password
The username must be lowercase and contain only letters, numbers, underscores, and hyphens.
APP OPTIONS
SKIP CUSTOMISATION
BACK
NEXT


Raspberry Pi Imager v2.0.6
Customisation: Choose Wi-Fi
Storage
Customisation
Hostname
Localisation
User
Wi-Fi
Remote access
Raspberry Pi Connect
Writing
Done
SECURE NETWORK
OPEN NETWORK
SSID:
Network name
Password:
Network password
Confirm password:
Re-enter password
Hidden SSID
APP OPTIONS
SKIP CUSTOMISATION
BACK
NEXT


Raspberry Pi Imager v2.0.6
Storage
Customisation
Hostname
Localisation
User
Wi-Fi
Remote access
Raspberry Pi Connect
Writing
Done
Customisation: SSH authentication
Configure SSH access
Enable SSH
Learn about SSH
Authentication mechanism:
Use password authentication
Use public key authentication
APP OPTIONS
SKIP CUSTOMISATION
BACK
NEXT


Raspberry Pi Imager v2.0.6
Storage
Customisation
Hostname
Localisation
User
Wi-Fi
Remote access
Raspberry Pi Connect
Writing
Done
Customisation: Raspberry Pi Connect
Sign in to receive a token and enable Raspberry Pi Connect
Enable Raspberry Pi Connect
What is Raspberry Pi Connect?
APP OPTIONS
SKIP CUSTOMISATION
BACK
NEXT


Raspberry Pi Imager v2.0.6
Device
OS
Storage
Customisation
Hostname
Localisation
User
Wi-Fi
Remote access
Raspberry Pi Connect
Write image
Review your choices and write the image to the storage device
Summary
Device: Raspberry Pi Zero 2 W
Operating system: Raspberry Pi OS Lite (32-bit)
Storage: Generic STORAGE DEVICE USB Device
Customisations to apply:
• Hostname configured
• Localisation configured
• User account configured
• Wi-Fi configured
• SSH enabled
APP OPTIONS
BACK
WRITE

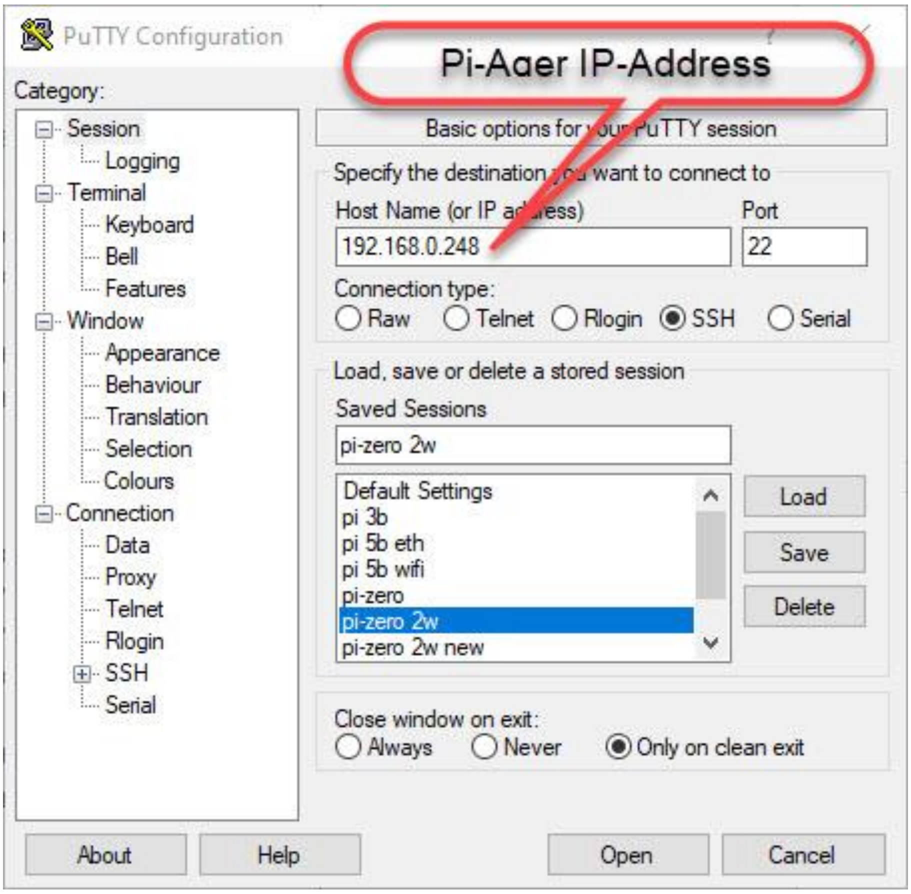

Raspberry Pi Imager v2.0.6
OS
Storage
Customisation
Hostname
Localisation
User
Wi-Fi
Remote access
Raspberry Pi Connect
Writing
Write complete!
Your choices:
Device: Raspberry Pi Zero 2 W
Operating system: Raspberry Pi OS Lite (32-bit)
Storage: Generic STORAGE DEVICE USB Device
Customisations applied:
✓ Hostname configured
✓ Localisation configured
✓ User account configured
✓ Wi-Fi configured
✓ SSH enabled
The storage device was ejected automatically. You can now remove it safely.
APP OPTIONS
WRITE ANOTHER
FINISH

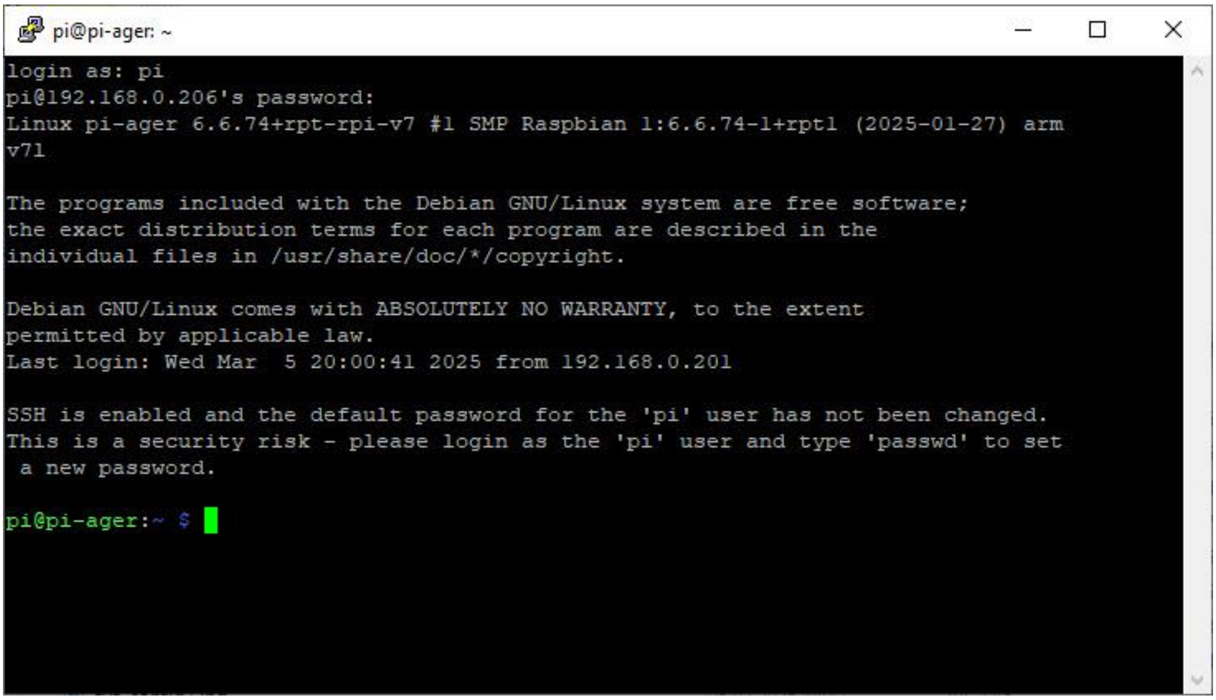
Put your SD-Card in your Raspberry Pi and power on your Pi device.
Login via SSH (e.g. use PuTTY) or connect a HDMI monitor and USB keyboard to your Raspberry device and continue with the setup as described below.

- Download and install PuTTY from https://www.putty.org/

- Optional download and install WinSCP from https://winscp.net/eng/index.php

- Start PuTTY to connect to Pi-Ager:





- Download system build scripts from github:
  wget -O pi-ager-build-system.sh -nv https://github.com/Tronje-the-Falconer/Pi-Ager/raw/refs/heads/resources/anleitungen/pi-ager-build-system.sh

  wget -O pi-ager-finalize-build.sh -nv https://github.com/Tronje-the-Falconer/Pi-Ager/raw/refs/heads/resources/anleitungen/pi-ager-finalize-build.sh

  wget -O setup-wifi-ap.sh -nv https://github.com/Tronje-the-Falconer/Pi-Ager/raw/refs/heads/resources/anleitungen/setup-wifi-ap.sh

  chmod +x *.sh

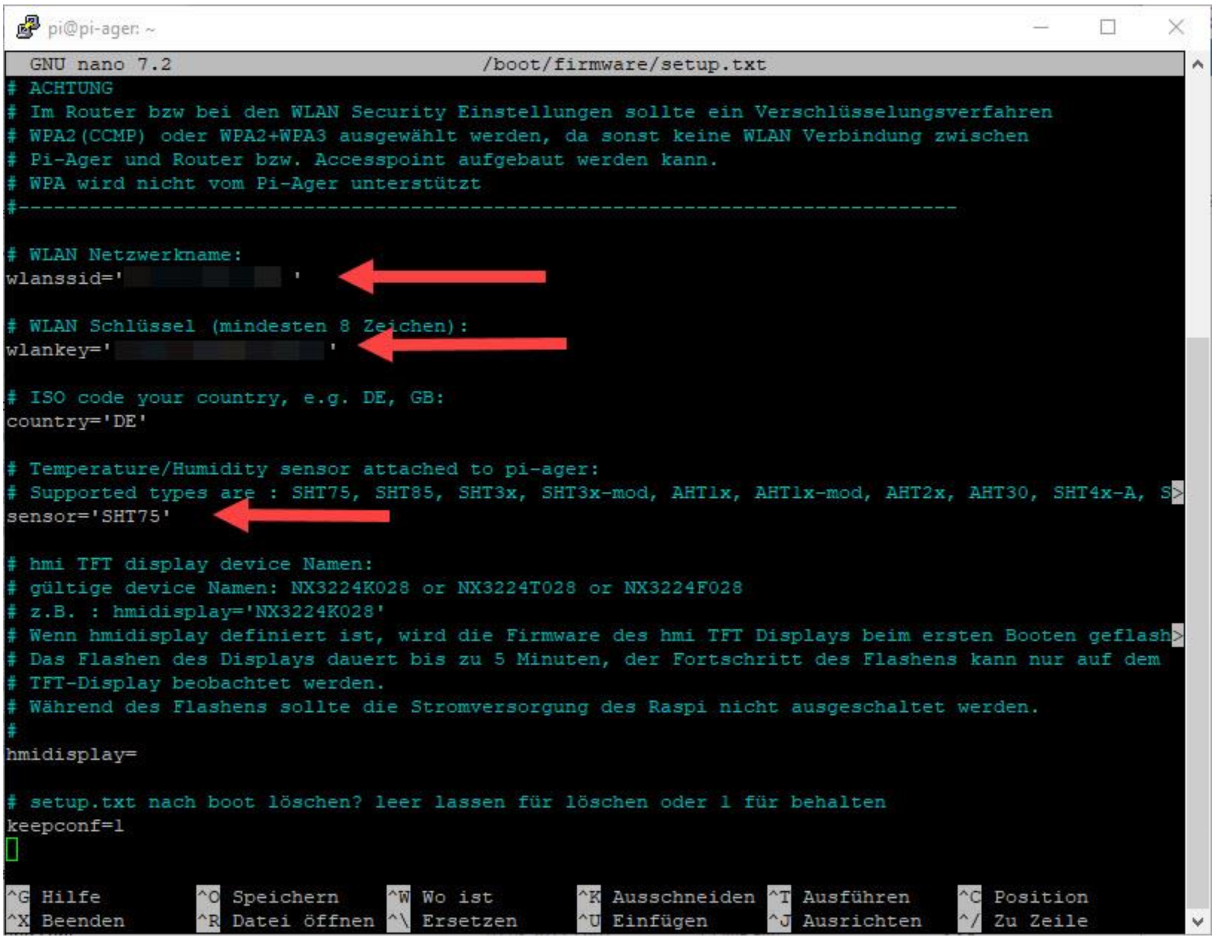
  Start now first phase of Pi-Ager System setup:
  sudo ./pi-ager-build-system.sh

  You can now follow the system setup, that is divided in 3 steps with rebooting between the steps. Restart the script after rebooting to continue the system setup :

  sudo ./pi-ager-build-system.sh

  After step 2 you are informed to edit the file /boot/firmware/setup.txt that contains information about passwords, WLAN setup parameter, temperature/humidity sensor to use and optional HMI display version connected to your Pi-Ager:

  sudo nano /boot/firmware/setup.txt



After saving setup.txt the last step initializes the Pi-Ager system with data from setup.txt by starting the script pi-ager-finalize-build.sh:

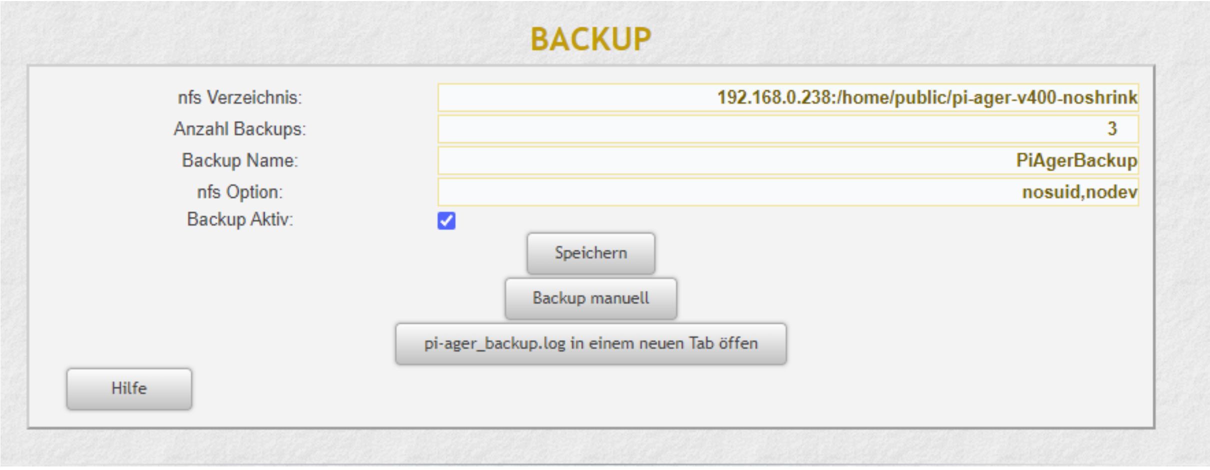
sudo ./pi-ager-finalize-build.sh

The system is then rebooted and ready to start the Pi-Ager. Open your browser window and enter the IP-Address of the Pi-Ager in the address line.

How to generate and deploy Pi-Ager Images

There are some steps to do for generating and deploying a new image from a running Pi-Ager system:

1. Setup a NFS Server to store a new image. The best way is using a Raspi 4B or 5 equipped with an USB Memory stick with 128GB or more. Using RPi 5 it is now possible to add a NVME SSD so that there will be enough space for storing one or more images.
   How to setup a NFS Server :
   https://www.elektronik-kompendium.de/sites/raspberry-pi/2007061.htm
2. On the Pi-Ager setup and start an Image backup on the ADMIN page. For example :



3. Next start a terminal on the Pi-Ager and start the pi-ager_image.sh script with option -l: (lowercase 'L')

```
pi@pi-ager: /usr/local/bin

pi@pi-ager:~ $ cd /usr/local/bin/
pi@pi-ager:/usr/local/bin $ sudo pi-ager_image.sh -l
Backup ist inaktiv. Pi-Ager_image wird gestartet!
überprüfe ob der NFS-Server vorhanden ist.
Checking...
192.168.0.238:/home/public/pi-ager-v400-noshrink ist vorhanden
überprüfe ob Backup Pfad vorhanden ist.
Checking ...
/home/nfs/public ist vorhanden
überprüfe ob NFS Mount point definiert ist.
Checking ...
/home/nfs/public ist definiert
überprüfe ob NFS Mount point im file system angelegt ist.
Checking...
/home/nfs/public ist angelegt
NFSVOL=192.168.0.238:/home/public/pi-ager-v400-noshrink
NFSPATH=/home/nfs/public
NFSMOUNT=/home/nfs/public
umount: /home/nfs/public: not mounted.
hänge NFS-Volume 192.168.0.238:/home/public/pi-ager-v400-noshrink ein
mount with options: nosuid,nodev
mount NFS-Volume 192.168.0.238:/home/public/pi-ager-v400-noshrink successfull. Image script continues
.
Load last backup file.
Backup path is  /home/nfs/public
Backup file is /home/nfs/public/PiAgerBackup_2024-02-12-103529.img
Backup path with file is /home/nfs/public/PiAgerBackup_2024-02-12-103529.img
Source File = /home/nfs/public/PiAgerBackup_2024-02-12-103529.img
do_copy     = false
my_image    = false
Using /home/nfs/public/PiAgerBackup_2024-02-12-103529.img as source and target
####################################################################################
parted_output = 2:541065216B:15931539455B:15390474240B:ext4::;
partnum = 2
partstart = 541065216
loopback = /dev/loop0
####################################################################################
parted_output_boot = 1:4194304B:541065215B:536870912B:fat32::lba;
partnum_boot = 1
partstart_boot = 4194304
loopback_boot = /dev/loop1
####################################################################################
mount directory is /tmp/tmp.SfM3akt9CW
2024-02-12 10:39:10 URL:https://raw.githubusercontent.com/Tronje-the-Falconer/Pi-Ager/entwicklung/boo
t/firmware/setup.txt [3281/3281] -> "setup.txt" [1]
setup.txt copied to /tmp/tmp.SfM3akt9CW/boot/
cmdline.txt modified, added init=/usr/lib/raspberrypi-sys-mods/firstboot
rematch1 /tmp/
rematch2 tmp.SfM3akt9CW
change rootdir is tmp.SfM3akt9CW
/tmp
pi-ager_main.service disabled, will be enabled during setup_pi-ager.sh
Created symlink /etc/systemd/system/multi-user.target.wants/setup_pi-ager.service → /lib/systemd/system/setup_p
i-ager.service.
setup_pi-ager service enabled
Running in chroot, ignoring command 'is-active'
hostname changed to pi-ager
unmount /tmp/tmp.SfM3akt9CW/boot
unmount /tmp/tmp.SfM3akt9CW
Detaching loop devices from /home/nfs/public/PiAgerBackup_2024-02-12-103529.img
The image /home/nfs/public/PiAger_image_2024-02-12-103940.img was successfully created.
pi@pi-ager:/usr/local/bin $
```

4. Zip and deploy your new image.